

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular Winter Examination – 2024

Course: B.Tech Branch : Electronics & Computer Engg/Electronics & Computer Science

Semester :VII

Subject Code & Name: BTECOE705B Cyber Security & Block chain Technology

Max Marks: 60

Date: 21/02/2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks				
Q. 1	Objective type questions. (Compulsory Question)		12				
1	Cryptanalysis is used to _____.	2	1				
	<table border="1"> <tr> <td>a. find some insecurity in a cryptographic scheme</td> <td>b. to increase the speed</td> <td>c. to encrypt the data</td> <td>d. to make new ciphers</td> </tr> </table>	a. find some insecurity in a cryptographic scheme	b. to increase the speed	c. to encrypt the data	d. to make new ciphers		
a. find some insecurity in a cryptographic scheme	b. to increase the speed	c. to encrypt the data	d. to make new ciphers				
2	Which of the following algorithm does not belong to symmetric key ciphers?	2	1				
	<table border="1"> <tr> <td>a. Rc4</td> <td>b. DES</td> <td>c. Diffie-Hellman</td> <td>d. AES</td> </tr> </table>	a. Rc4	b. DES	c. Diffie-Hellman	d. AES		
a. Rc4	b. DES	c. Diffie-Hellman	d. AES				
3	Which of the following is incorrect about symmetric key cryptography?	1	1				
	<table border="1"> <tr> <td>a. It uses common key for both encryption and decryption</td> <td>b. It is a secret key cryptography</td> <td>c. It uses single fixed length key for each algorithm</td> <td>d. Symmetric key can be combined with asymmetric key to achieve more security</td> </tr> </table>	a. It uses common key for both encryption and decryption	b. It is a secret key cryptography	c. It uses single fixed length key for each algorithm	d. Symmetric key can be combined with asymmetric key to achieve more security		
a. It uses common key for both encryption and decryption	b. It is a secret key cryptography	c. It uses single fixed length key for each algorithm	d. Symmetric key can be combined with asymmetric key to achieve more security				
4	Electronic Code Book process is used in _____.	1	1				
	<table border="1"> <tr> <td>a. caesar cipher</td> <td>b. antisymmetric cipher</td> <td>c. block cipher</td> <td>d. stream cipher</td> </tr> </table>	a. caesar cipher	b. antisymmetric cipher	c. block cipher	d. stream cipher		
a. caesar cipher	b. antisymmetric cipher	c. block cipher	d. stream cipher				
5	_____ is an example of asymmetric ciphers.	3	1				

	a. Block cipher	b. RSA	c. AES	d. Advanced cryptography		
6	What type of algorithm does AES encryption use?				3	1
	a. Stream cipher	b. Symmetric block cipher	c. Asymmetric caesar cipher	d. DES encryption		
7	What is the first block of a Blockchain called?				5	1
	a. Seed Block	b. Origin Block	c. Genesis Block	d. Root Block		
8	What is the unit structure of a Blockchain called?				5	1
	a. Node	b. Transaction	c. Block	d. Record		
9	When a hash function is used to provide message authentication, the hash function value is referred to as				4	1
	a. Message Field	b. Message Digest	c. Message Score	d. Message Leap		
10	Message authentication code is also known as				4	1
	a. key code	b. hash code	c. keyed hash function	d. message key hash function		
11	Which of the following actions compromise cyber security?				1	1
	a. Vulnerability	b. Attack	c. Threat	d. Exploit		
12	What is the primary purpose of Blockchain technology?				5	1
	a. To replace all traditional databases	b. To provide a secure and transparent record of transactions	c. To create cryptocurrencies	d. To centralize data storage		
Q. 2	Solve the following.					12
A)	Compare stream cipher with block cipher with example.				2	6
B)	Elaborate working of Blockchain technology.				5	6
Q.3	Solve the following.					12

A)	With the help of block diagram AES algorithm in detail.	3	6
B)	What is data integrity, non-repudiation and authentication?	1	6
Q. 4	Solve Any Two of the following.		12
A)	Describe Elliptic Curve Cryptography algorithm.	4	6
B)	Elaborate RSA algorithm with example.	2	6
C)	What is HMAC?	2	6
Q.5	Solve Any Two of the following.		12
A)	Describe SSL in detail.	3	6
B)	Explain the encryption and decryption of DES algorithm in detail.	3	6
C)	Compare public Blockchain with private Blockchain.	5	6
Q. 6	Solve Any Two of the following.		12
A)	Identify the possible threats for RSA algorithm and list their counter measures.	3	6
B)	Write a note on classical cryptosystems.	1	6
C)	How firewalls provides security?	4	6
	*** End ***		